Predicting Unemployment Using Economic Indicators: A Machine Learning Approach

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Introduction

Unemployment is one of the most important indicators of economic health and labour market stability. It measures the proportion of the labour force that is actively seeking employment but unable to find work. High unemployment can signal economic inefficiencies, reduced productivity, and broader social challenges such as income inequality and declining living standards. For this reason, understanding the factors that influence unemployment is a key concern for economists and policymakers.

Several macroeconomic indicators can influence unemployment levels. Economic growth affects job creation, inflation reflects price stability and overall economic conditions, labour force participation indicates the share of the population engaged in the labour market, and population growth captures demographic changes that may affect labour supply. However, the relationships between these variables and unemployment are often complex and vary across different economies.

This study uses data from the **World Bank's World Development Indicators** to analyse unemployment trends across selected countries from **1991 to 2024**. By combining exploratory data analysis with machine learning models, the project aims to identify which economic indicators most strongly influence unemployment and evaluate how effectively these variables can predict unemployment levels.

Problem Statement and Objectives

Unemployment trends vary significantly across countries due to differences in economic performance, labour market structures, and demographic conditions. While macroeconomic indicators such as GDP growth, inflation, labour force participation rate (LFPR), and population growth are often associated with labour market outcomes, it is not always clear which variables have the strongest influence on unemployment.

The main objective of this study is to analyse the relationship between unemployment and selected macroeconomic indicators. Specifically, the study aims to examine how these indicators influence unemployment patterns and evaluate the ability of machine learning models to predict unemployment levels based on these variables.

Dataset and Variables

The dataset used in this study was obtained from the **World Bank World Development Indicators database**. The analysis covers the period **1991–2024** and includes data from ten countries: the United States, Japan, France, the United Kingdom, Canada, India, Brazil, Mexico, Indonesia, and South Africa.

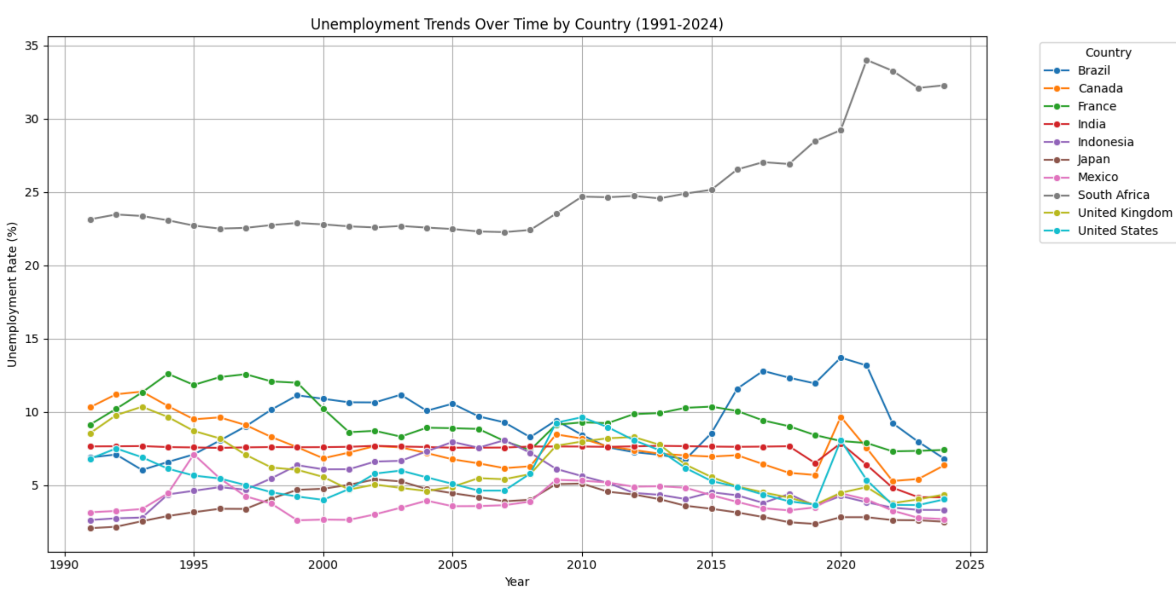
The study uses one dependent variable and four independent variables.

The dependent variable is the unemployment rate (% of the labour force). The independent variables include GDP growth, inflation, labour force participation rate (LFPR), and population growth. These variables represent key economic and demographic factors that may influence labour market conditions.

Methodology

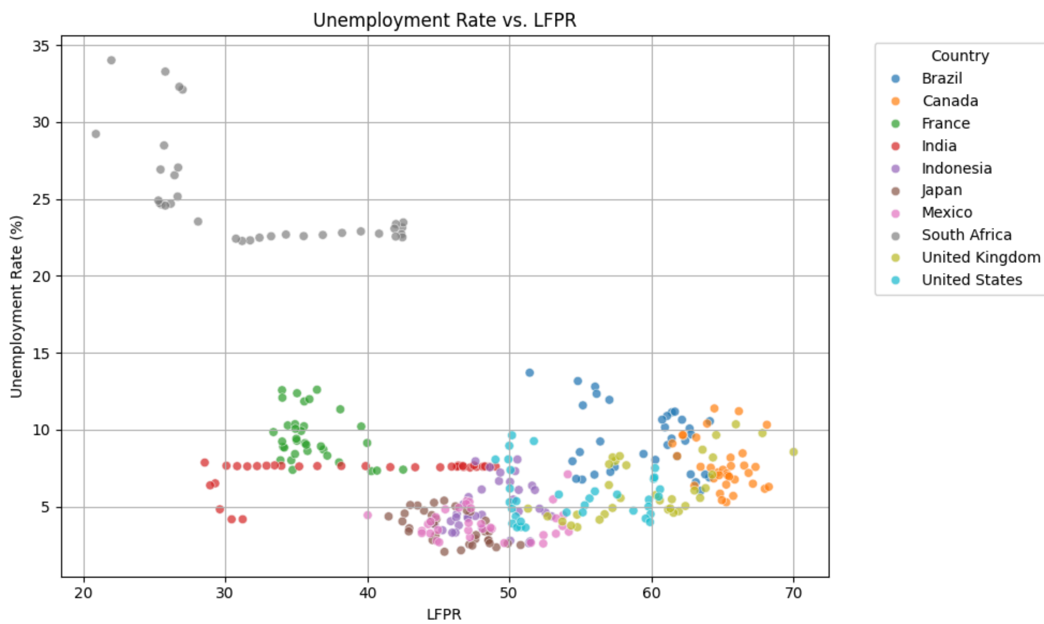
The methodology involved three main steps: data preparation, exploratory data analysis, and machine learning modelling.

First, the datasets for each indicator were collected and combined into a single dataset using country and year as common identifiers. The data was then cleaned to remove missing values and ensure consistent formatting. Exploratory data analysis (EDA) was conducted to understand patterns in the data and examine relationships between unemployment and the selected economic indicators.

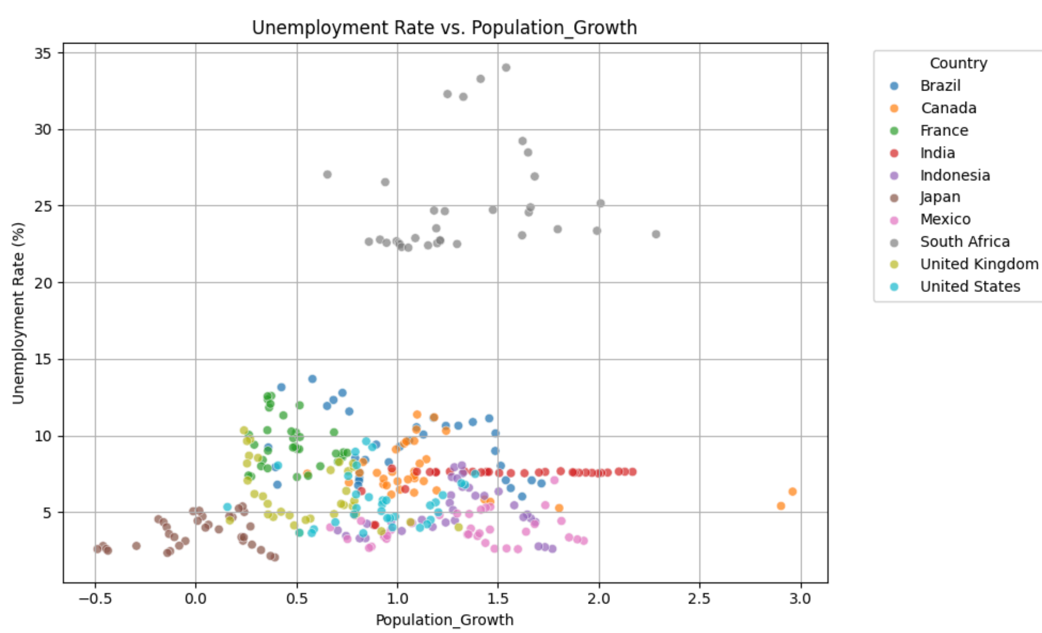


This graph shows how unemployment levels have evolved across different countries over time. The trends highlight significant differences between economies, with some countries maintaining relatively stable unemployment while others experience higher and more volatile rates.

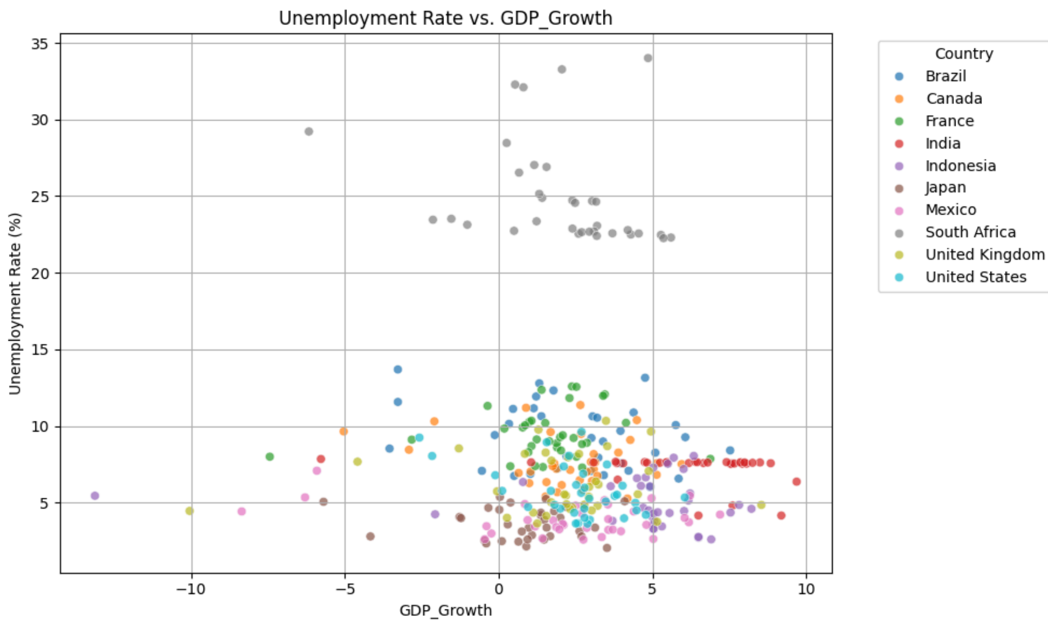
Scatter plots were also used to explore relationships between unemployment and individual economic indicators.



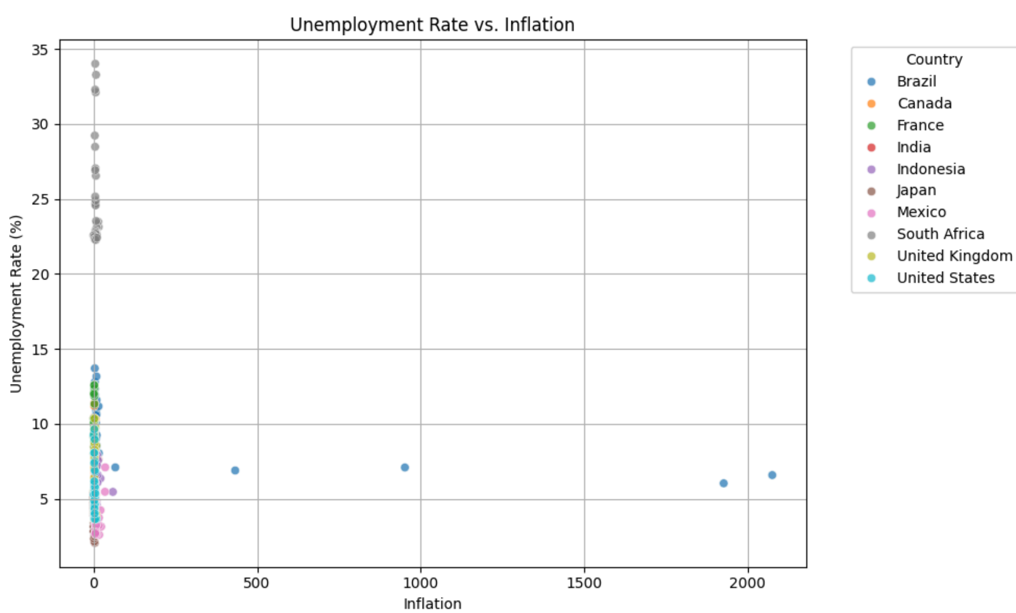
The scatter plot suggests that labour force participation plays an important role in unemployment patterns, with higher participation levels often associated with lower unemployment.



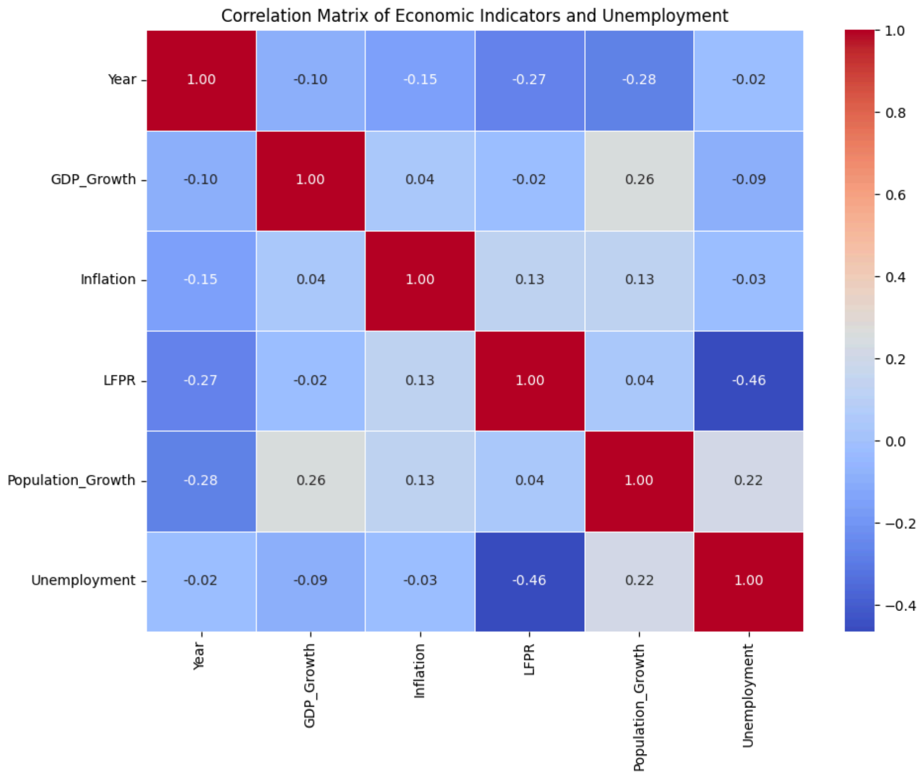
This visualisation shows that population growth alone does not strongly determine unemployment levels, indicating that demographic changes interact with other economic factors.



The relationship between GDP growth and unemployment reflects the economic principle that stronger economic growth generally leads to greater employment opportunities.



The inflation plot indicates a relatively weak direct relationship between inflation and unemployment within the dataset.



The correlation heatmap summarises the relationships between all variables and highlights which indicators show stronger associations with unemployment.

Machine Learning Results

Several machine learning models were applied to analyse the dataset, including Decision Tree Regressor, Random Forest Regressor, K-Nearest Neighbours Regressor, and Support Vector Regressor. The models were evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score.

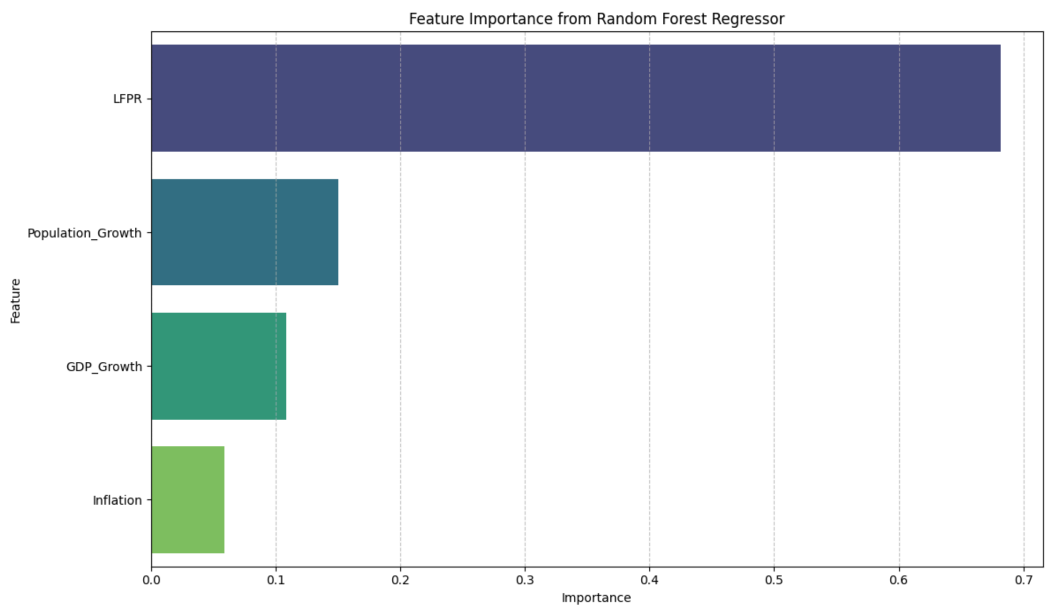
Model Evaluation Results:

	Model	MSE	RMSE	R-squared
0	Decision Tree Regressor	7.332270	2.707816	0.583747
1	Random Forest Regressor	5.172043	2.274213	0.706383
2	K-Nearest Neighbors Regressor	7.188241	2.681090	0.591923
3	Support Vector Regressor	16.471743	4.058540	0.064898

The model evaluation results show that the Random Forest Regressor performed the best among the tested models, achieving the lowest Mean Squared Error (MSE = 5.17) and Root Mean Squared Error (RMSE = 2.27), along with the highest R^2 score of 0.71. In comparison, the Decision Tree Regressor achieved an R^2 score of 0.58, while the K-Nearest Neighbours model produced an R^2 score of 0.59. The Support Vector Regressor performed poorly with an R^2 score of 0.06, indicating that it was not well-suited for this dataset.

This suggests that the Random Forest model was most effective at capturing the relationships between the economic indicators and unemployment.

To better understand the influence of each variable on unemployment predictions, feature importance analysis was conducted using the Random Forest model.



Feature importance analysis using the Random Forest model reveals that Labour Force Participation Rate (LFPR) is the most influential variable, accounting for approximately 67% of the model's predictive importance. Population growth accounts for about 15%, while GDP growth accounts for about 11% of the model's explanatory power. Inflation shows the lowest importance (around 6%), suggesting it plays a relatively weak role in predicting unemployment in the dataset.

Key Findings and Recommendations

The analysis reveals several important insights. First, unemployment patterns vary widely across countries, reflecting differences in economic structures and labour market dynamics. Second, labour force participation appears to have the strongest relationship with unemployment levels, suggesting that policies encouraging workforce participation could significantly influence labour market outcomes.

Economic growth also plays an important role in shaping unemployment trends, highlighting the importance of sustainable economic expansion for job creation. Population growth shows a moderate influence, indicating that demographic pressures may affect labour markets over time.

Based on these findings, policymakers should focus on encouraging workforce participation through education, training, and flexible employment opportunities. Supporting economic growth through investment and innovation can also contribute to improved employment outcomes. Additionally, using data-driven approaches such as machine learning can help governments better anticipate changes in labour market conditions and design more effective policies.